

Comparison of the effects of whole and juiced fruits and vegetables on enamel demineralisation in situ.

We compared the effect on enamel demineralisation in situ of both whole and juiced fruits and vegetables. Volunteers wore removable mandibular appliances carrying pre-demineralised human enamel slabs and consumed one of the test foods 7 times a day for 10 days. The test foods were apples, oranges, grapes, carrots, and tomatoes, consumed either whole (sugars located intrinsically) or as a juice (extrinsic or free sugars). Raisins containing 64% sugars, but intrinsic by definition, were also studied. The mineral profile of the enamel slabs was studied before and after the test period using transverse microradiography and showed further demineralisation for all test foods, irrespective of the form of consumption. Significant demineralisation was also observed with raisins. No significant differences were found between the solid and juiced foods. In conclusion, sugars present intrinsically on consumption had a similar demineralising potential as free sugars and could not be considered less cariogenic. Copyright © 2011 S. Karger AG, Basel.